

Flow Sight

A Real-Time Data Engineering Pipeline for Traffic Analytics

Fundamentals of Data Engineering Project

Team 7

Members:

J. Nandini – 2420030262
S. Sai Saranya – 2420030266
B. Sai Charan – 2420030364

Faculty Guide

Dr. N. Sirisha
Associate Professor

Project Overview



01 Problem, Purpose & Objectives

02 Literature Survey

03 Research Gap & Proposed Solution

04 System Architecture

05 Data Generation, Validation & Transformation

06 Expected Output & Conclusion

Problem Statement & Project Purpose

Problem Statement

Traffic sensors can generate continuous streams of vehicle-count and speed data. Raw events may contain missing values, duplicate readings or invalid measurements. Manual handling becomes difficult as the data grows, creating a need for an automated pipeline that can ingest, validate, transform and organize the stream.

Project Purpose

FlowSight converts continuously generated traffic events into clean, structured and analytics-ready information. The processed data is used to identify peak traffic hours, frequently congested roads, average speeds, vehicle volume and data-quality issues.

Core idea: Build a reliable data pipeline — not a machine-learning traffic prediction system.

Generate → Stream → Validate → Transform → Store → Analyze → Visualize

Project Objectives

01

Generate traffic data

02

Stream the data in real time

03

Validate and clean raw traffic records.

04

Transform raw events into analytics-ready information.

05

Congested Road Identification

06

Real-Time Traffic Monitoring

Literature Survey

In the literature survey, we reviewed previous work in three connected areas: traffic and intelligent transportation data, real-time event streaming, and modern data-pipeline architectures. The review helped identify the need to handle continuously generated events through separate ingestion, processing, storage and analytics stages.

Traffic & Transport Data

Traffic systems generate continuous data from sensors, vehicles and monitoring sources. Research emphasizes the need for scalable collection and analysis of these high-volume data streams.

Real-Time Event Streaming

Streaming systems capture events from sources, move them continuously through the pipeline and support downstream processing. Kafka is widely used for this event-driven approach.

Pipeline Architecture

Modern architectures separate source generation, ingestion, processing, storage and analytics so each stage has a clear responsibility and can be improved independently.

Result of the survey: FlowSight combines these ideas into a manageable end-to-end learning project.

Literature Survey — Related Work

Study	Area	Key takeaway for FlowSight
Raptis & Passarella (2023) A Survey on Networked Data Streaming With Apache Kafka https://doaj.org/article/0bf1606df61b4c54a2efd970704b5c89?utm_source=chatgpt.com	Kafka / networked streaming	Shows Kafka's relevance across data, cyber-physical and streaming applications; supports using Kafka as the ingestion layer.
Kreps, Narkhede & Rao (2011) Kafka: a Distributed Messaging System for Log Processing https://kafka.apache.org/community/books_and_papers/?utm_source=chatgpt.com	Distributed event processing	Establishes Kafka as a publish/subscribe system for high-throughput event movement between producers and consumers.
Singh (2024) Real-Time Data Processing ... Traffic Analysis https://www.ijirmps.org/research-paper.php?id=230767&utm_source=chatgpt.com	Real-time processing / traffic	Highlights the value of real-time processing frameworks and low-latency analytics for traffic-related applications.

Research Gap & Proposed Solution

Existing Approaches

Many systems are designed for large-scale production use, use multiple distributed tools, or focus strongly on traffic prediction and machine learning.

Identified Gap

For learning Fundamentals of Data Engineering, students need a smaller project that still demonstrates the complete journey of continuously arriving data through a realistic pipeline.

FlowSight Solution

FlowSight provides a beginner-friendly but complete architecture:

Python traffic simulator



Kafka event ingestion



Validation & transformation



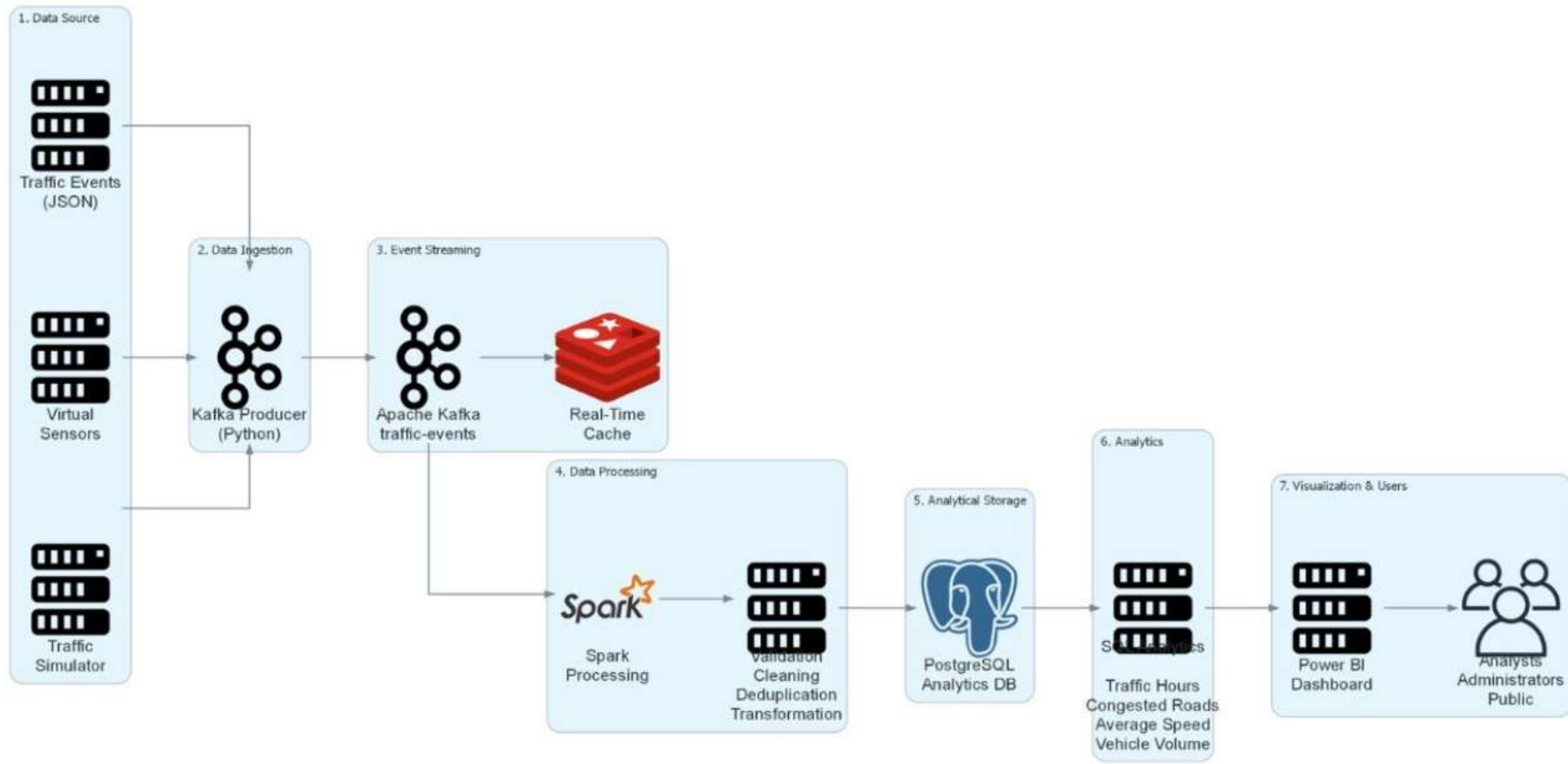
PostgreSQL historical storage



SQL analytics & Power BI visualization

Focus: demonstrate Data Engineering fundamentals clearly and practically.

System Architecture:



FlowSight - Real-Time Traffic Analytics

Technology Selection

Python

Easy way to simulate continuously arriving traffic events and implement processing logic.

Apache Kafka

Acts as the event-streaming layer between the producer and consumer.

PostgreSQL

Stores clean structured records for historical queries and analysis.

SQL

Calculates averages, totals, peak periods and road-level traffic patterns.

Power BI

Presents processed information through dashboards and KPIs.

Data Generation

```
{  
  "sensor_id": "S003",  
  "road_name": "Hitech City Road",  
  "timestamp": "2026-08-06 08:35:20",  
  "vehicle_count": 230,  
  "average_speed": 17  
}
```

Source Simulation

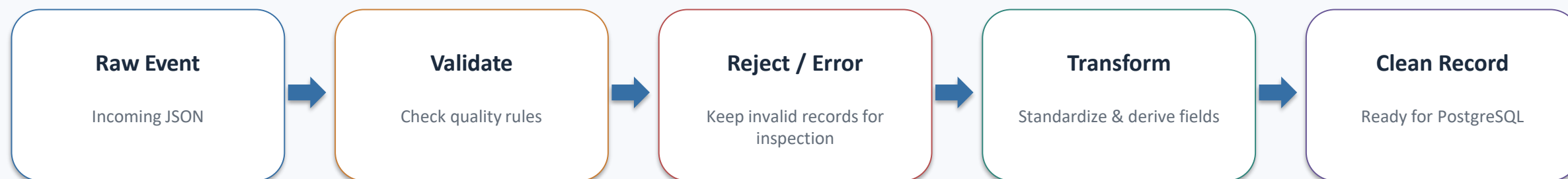
Python simulates multiple virtual traffic sensors and continuously creates events for different roads and times.

Why Simulated Data?

It allows the complete pipeline to be built and demonstrated without requiring physical traffic sensors or access to a live city traffic feed.

The event format is designed to be simple, structured and suitable for streaming as JSON.

Data Validation & Transformation



Validation Rules

- sensor_id required
- timestamp required
- vehicle_count ≥ 0
- $0 \leq \text{speed} \leq 150$
- duplicate events checked

Derived Fields

hour, day, month, year, weekday, weekend_flag and congestion_level are calculated from the raw event.

Example Rule

If vehicle_count > 200 AND average_speed < 20 km/h, classify the event as HIGH congestion.

Expected System Output

Peak Traffic Hours

Identify periods with the highest vehicle volume.

Congested Roads

Find roads with high volume and low average speed.

Average Speed

Compare road performance over time.

Vehicle Volume

Analyze total and average traffic counts.

Historical Trends

Compare traffic conditions across dates and periods.

Dashboard KPIs

Present results clearly through Power BI.

Final output: clean data + historical storage + meaningful traffic insights.

Conclusion

Complete Pipeline

Demonstrates the movement of data from source generation to ingestion, processing, storage and analytics.

Practical Learning

Combines Python, Kafka, PostgreSQL, SQL and Power BI in one connected Data Engineering workflow.

Future Ready

The architecture can later be extended with PySpark, Airflow, larger datasets or additional data sources.

FlowSight transforms continuously generated traffic events into clean, structured and analytics-ready information.

It focuses on the Fundamentals of Data Engineering lifecycle rather than machine-learning prediction.

Thank You